

Escherichia coli

Escherichia coli is a Gram-negative, rod-shaped bacterium and a member of the normal gut flora. It is the most common cause of community-acquired urinary tract infections, responsible for over 80% of cases. Key virulence factors include type 1 and P fimbriae (adhesion) and, in enterohemorrhagic strains (EHEC, e.g. O157:H7), Shiga toxin, which inhibits protein synthesis and can cause hemolytic uremic syndrome. Uncomplicated cystitis is typically treated with nitrofurantoin or trimethoprim-sulfamethoxazole. Resistance is increasingly mediated by extended-spectrum beta-lactamases (ESBL), which confer resistance to most beta-lactam antibiotics; carbapenems are used for ESBL-producing strains.